SAIDI and SAIFI Benchmark Comparison

Purpose. This note provides a transparent comparison of SAIDI and SAIFI provisions across the 12 most populous Indian states and selected international reference systems. For Indian states, the table distinguishes between explicit regulatory standards/targets, reporting requirements, and cases where no verified numeric benchmark has yet been identified. For international comparators, the figures shown are observed system performance rather than regulatory standards.

Indian states covered. Uttar Pradesh, Bihar, Maharashtra, West Bengal, Madhya Pradesh, Rajasthan, Tamil Nadu, Gujarat, Karnataka, Andhra Pradesh, Odisha and Telangana.

Master comparison table

Geography	Status of figure	SAIDI - original value	SAIFI - original value	Scope / coverage	Key methodological qualifications	Reference / source	Year
India	Range of available state-level regulatory benchmarks	1,200-14,400 minutes per consumer per year	40 to 300 interruptions/customer/year	Range derived from available numerical regulatory standards/targets identified from seven DISCOMs - AP (3), Rajasthan (3) and Tamil Nadu (1); not a national regulatory standard	Represents the minimum and maximum among identified state-level numerical benchmarks, not national minimum/ maximum performance. States without a verified numerical benchmark are excluded from the range.	Listed below	
United States	Actual observed performance	131.6 min/customer/year	1.065 times/year	U.S. distribution utilities; IEEE methodology	Figure shown excludes Major Event Days. EIA separately publishes all-events figures, which are materially higher.	EIA Electric Power Annual, 2024	
Japan	Actual observed performance	24 min/customer/year	0.13 interruptions/customer/year	Nationwide low-voltage customers	Reported total includes forced and planned interruption components. FY2024: SAIFI 0.11 forced + 0.02 planned; SAIDI 21 min forced + 3	OCCTO, FY2024	

					min planned.		
Germany	Actual observed performance	11.7 min/customer/year	Not reported in selected source	Electricity end customers covered by Bundesnetzagentur interruption reporting	Planned interruptions and force majeure excluded; only interruptions >3 min counted.	Bundesnetzagentur, 2024	
Singapore	Actual observed performance	0.32 min/customer/year	0.022 interruptions/customer/year	National electricity system	Compact city-state; useful as a reference point but structurally very different from an Indian state.	EMA, 2025	
Brazil	Actual observed performance	618 min/customer/year	4.9 interruptions/customer/year	Distribution system	Observed national distribution performance, not a regulatory standard.	PSR, 2024	
Argentina	Actual observed performance	906 min/customer/year	7.3 interruptions/customer/year	Distribution system	Observed national distribution performance, not a regulatory standard.	PSR, 2024	
Mexico	Actual observed performance	1068 min/customer/year	0.4 interruptions/customer/year	Distribution system	Observed national distribution performance, not a regulatory standard.	PSR, 2024	
China	Actual observed performance	**403** min/customer/year (6.71 h)	2.12 interruptions/customer/year	National, 10 kV customers	-	Regulator (NEA)	2024
Indonesia	Actual observed performance	**170** min/customer/year (2.84 h)	1.41 interruptions/customer/year	Rosseti group only; per point of supply	-	Utility annual report	2023

Benchmark National Average for India

The weighted benchmark is about 6,600 minutes (110 hours) SAIDI and about 152 interruptions SAIFI per consumer per year. Tamil Nadu makes up half the result.

Methodological notes

Annualisation. Where an Indian regulation specifies a quarterly, half-yearly or other non-annual standard, the annual value is a simple arithmetic conversion provided only to aid visual comparison. It does not represent the legally applicable compliance period.

Comparability. These figures are not strictly like-for-like. Indian entries principally describe regulatory standards, targets or reporting requirements, whereas the international entries describe observed performance. Jurisdictions also differ in their treatment of planned interruptions, major-event days / force majeure, minimum interruption duration, voltage levels, agricultural feeders and consumer coverage. These qualifications should remain visible alongside the numbers rather than being normalised away.

Interpretation. A blank or 'no verified numeric benchmark' entry is intentional. It indicates that the current first-pass review did not identify a defensible operative numeric standard in the core regulation/order reviewed. It should not be read as evidence that the regulator does not monitor SAIDI/SAIFI.